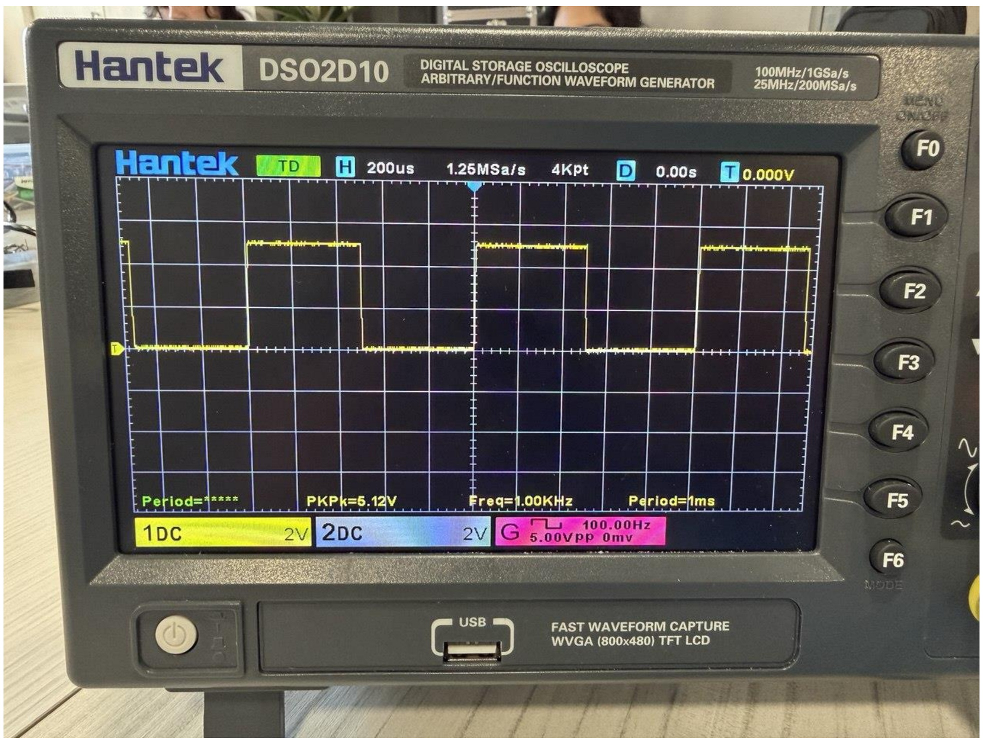
RLC Time Domain Analysis

ENGS152 Circuits Lab

28 February 2026

Step 1: Capacitor charge/discharge RC circuit

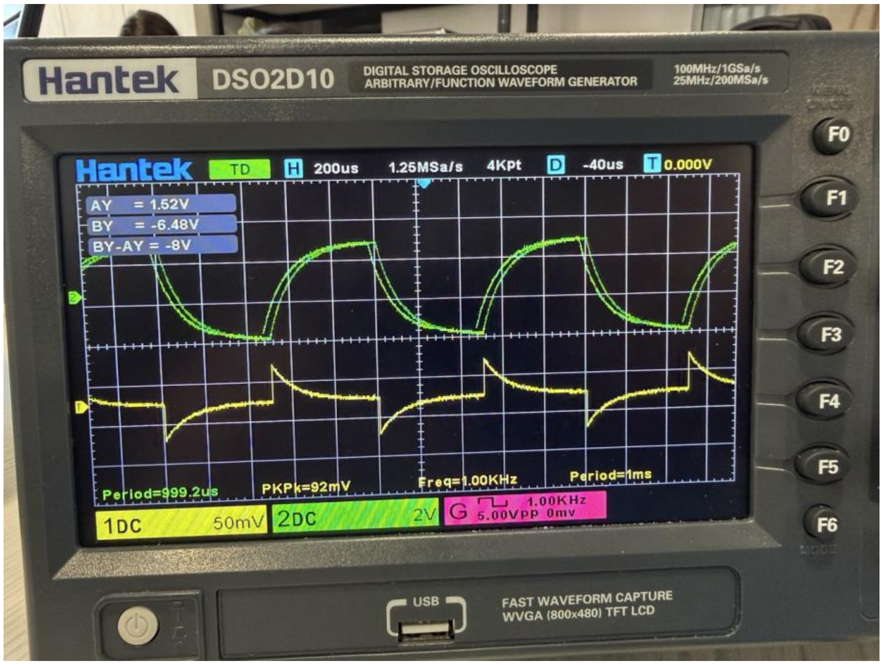


We started by generating square wave with frequency of 1 kHz using an oscilloscope. In the picture above you can see information about the Peak-to-Peak Voltage (PkPk), Frequency (Freq), and Period (Period). It is known the the capacitor is to be 100nF and the time constant 0f 0.1ms. From these variables the R value can be calculated for the RC circuit.

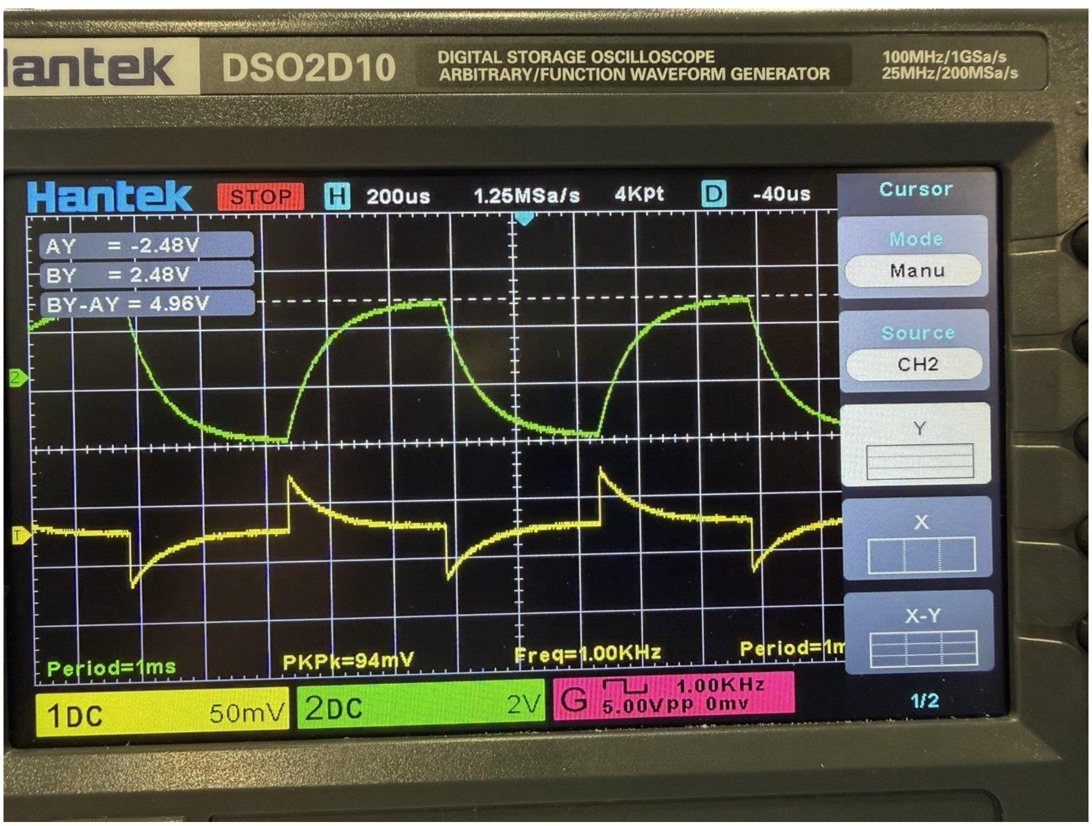
$$\tau = RC$$

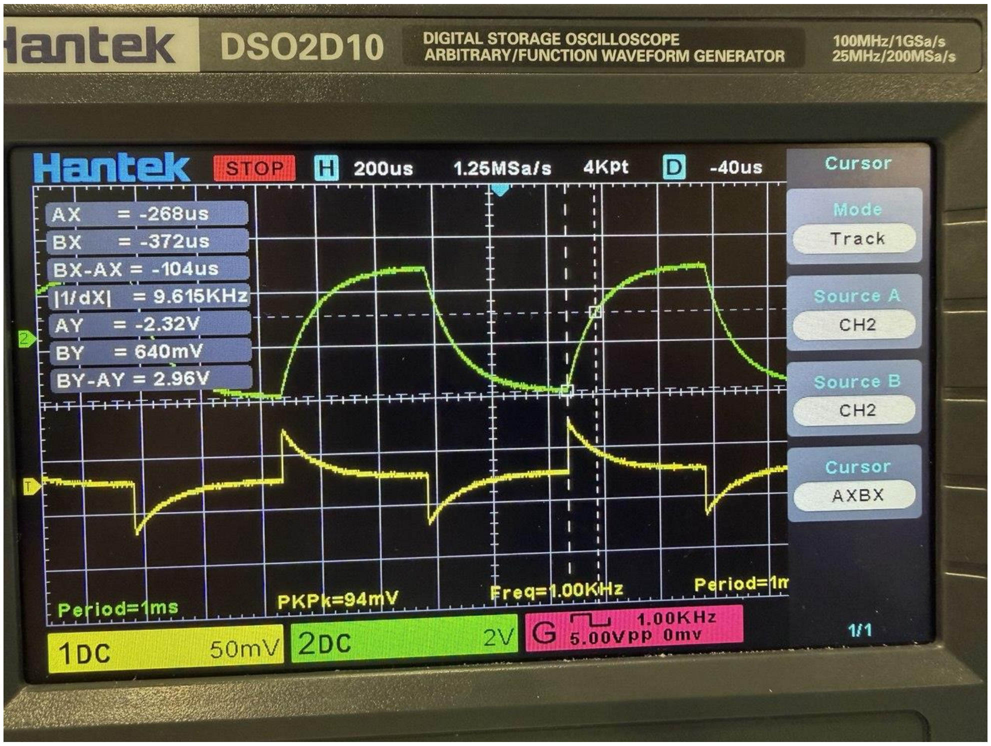
$$R = \frac{\tau}{R} = \frac{0.1ms}{100nF} = 1k\Omega$$

Voltage and current graphs are like those in Figure 9.1 and 9.3.



In the picture below with help of Cursor Y, we confirm that the voltage change is 5V. (Upper right side of the screen)





You can see that the cursor shows that voltage reaches ~63% of its final value in 1T = 0.1s time. 2.96V is almost 63% of 4.96V voltage change. (Keeping in mind errors).

Now as we substitute a constant resistor with a 10k potentiometer we can change the R value and observe how the time constant changes and the graph gets affected.

As we increase the R, the time constant τ, increases, capacitor charges slower and graph becomes more stretched.

As we decrease the R, the time constant τ, decreases and the capacitor charges faster so the graph becomes sharper.

(Lower R, Standard R tuning, Higher R tuning)

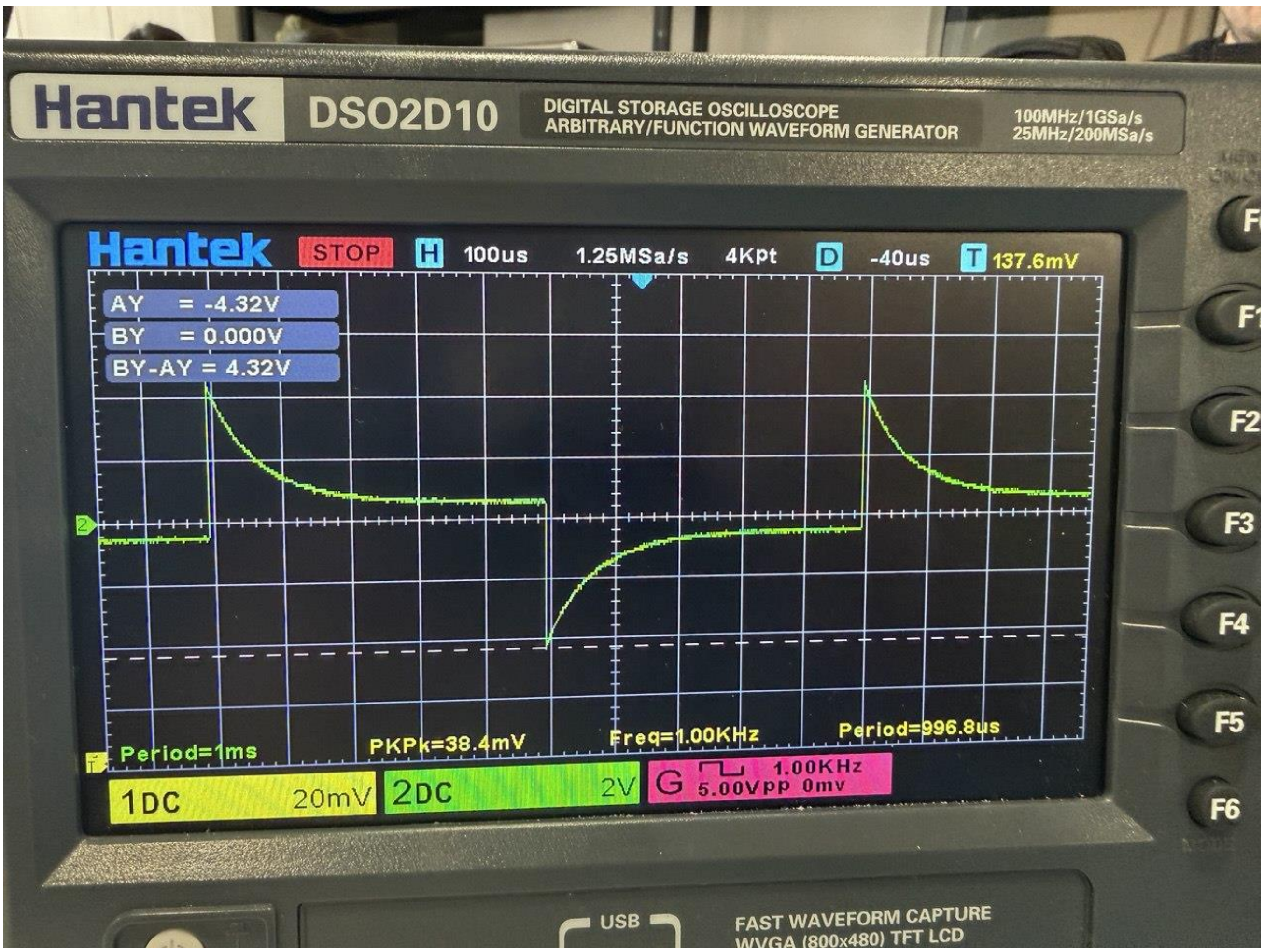
Step 2: Inductor energizing/de-energizing circuit

Square wave with frequency of 1 kHz has been generated again. The probe settings have been adjusted to their proper settings and the R value for the variables of H = 100mH, and time constant of 0.1ms has been calculated again.

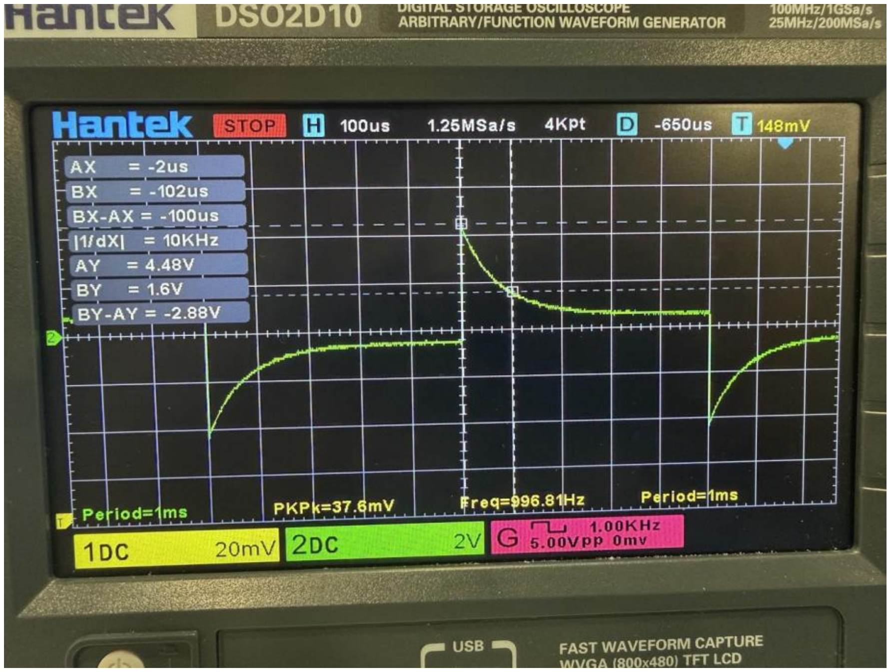
$$\tau = \frac{L}{R}$$

$$k = \frac{L}{\tau} = \frac{100mH}{0.1ms} = 1k\Omega$$

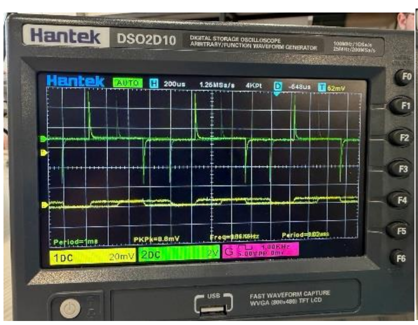
Below you can see both that the voltage graph of the corresponds to the graphs in figures 9.2 and 9.4, and that the voltage change is 4.32V (We were aiming for 5V).

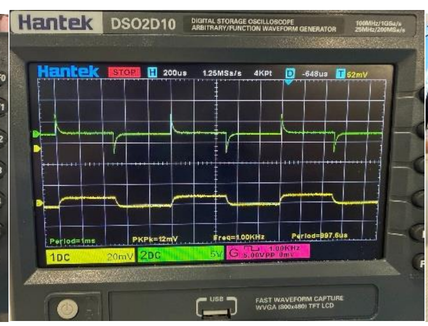


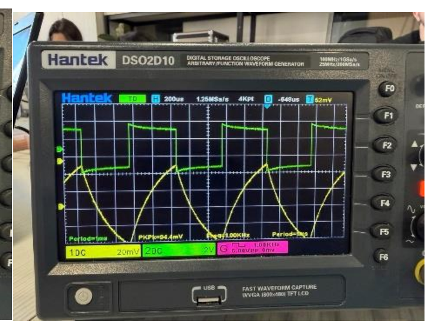
In the graph below it is visible that the voltage reaches 4.32 V – 2.88V = **1.44V** in 1T = 0.1ms time. Which is approximately the ~37% of its final value.



After substituting the constant resistor with a 10k potentiometer we can once again adjust the R value and affect the time constant.



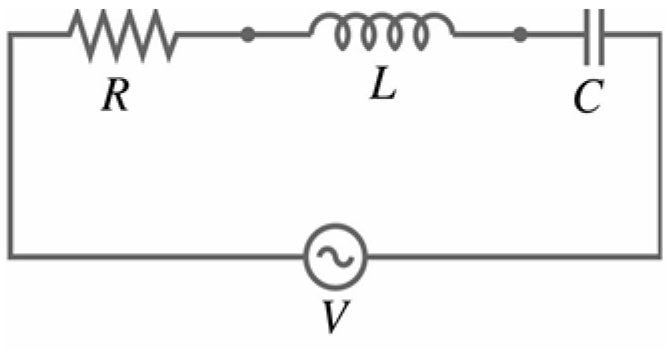




As we increase the R, the time constant τ, decreases, current rises/decays faster and the graph becomes steeper.

As we decrease the R, the time constant τ, increases and the current changes slower so the graph becomes more stretched.

Step 3: RLC series Circuit



In this part of the lab a square wave with frequency of 100 Hz is generated. The output voltage across the capacitor is measured. Capacitor of capacitance 100nF and inductor with inductance of 100mH is used, and the damping is controlled by a potentiometer by changing the resistance.

$$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{1}{c}i = 0$$

$$r^2 + \frac{R}{L}r + \frac{1}{LC} = 0$$

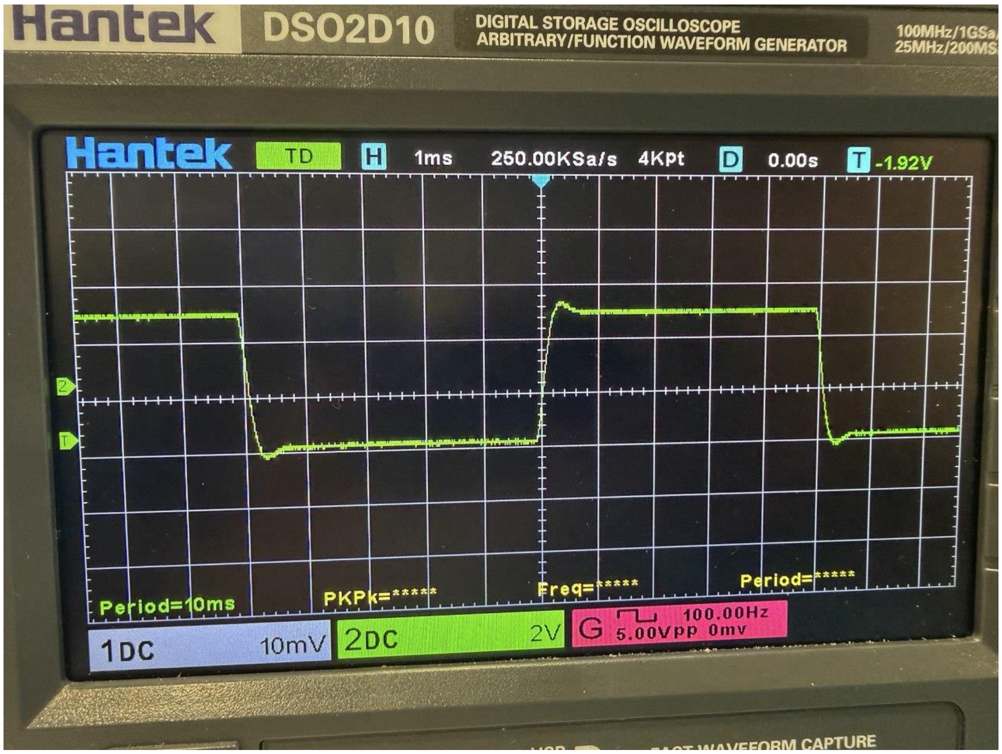
Critically damp when discriminant $\Delta = R^2 - \frac{4L}{C} = 0$

$$R = 2L\frac{1}{\sqrt{LC}} = 2\sqrt{\frac{L}{c}}$$

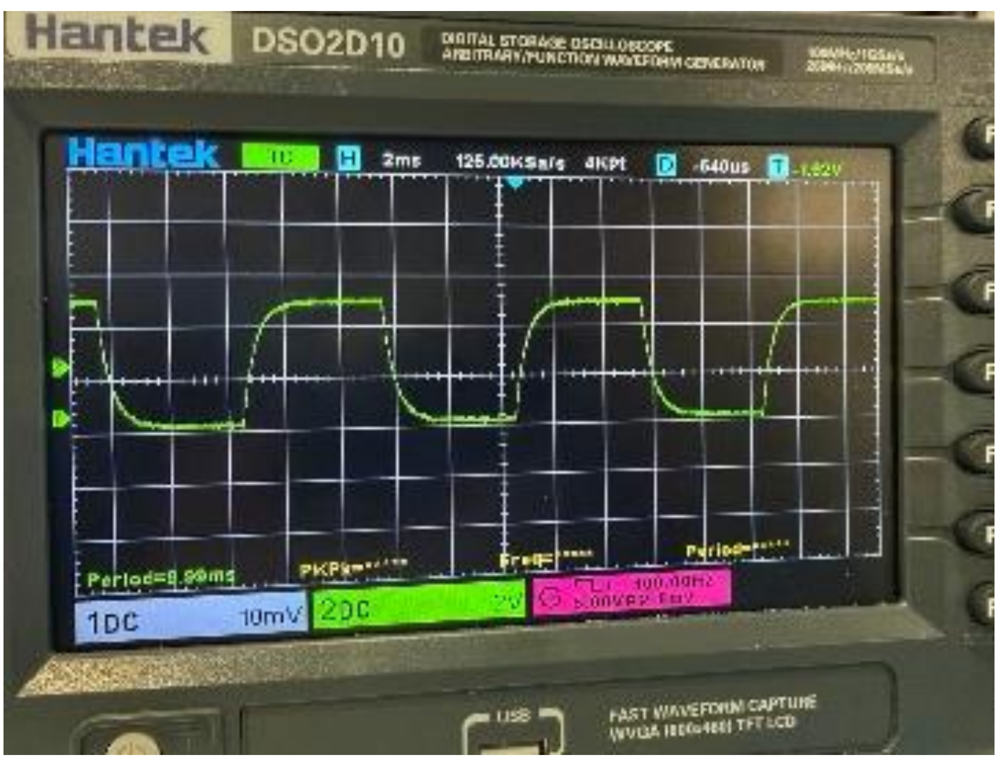
After inserting our parameters, we get the value of **2k*Ω*** for a critically damp response.

Below you can see the response at different damping settings.

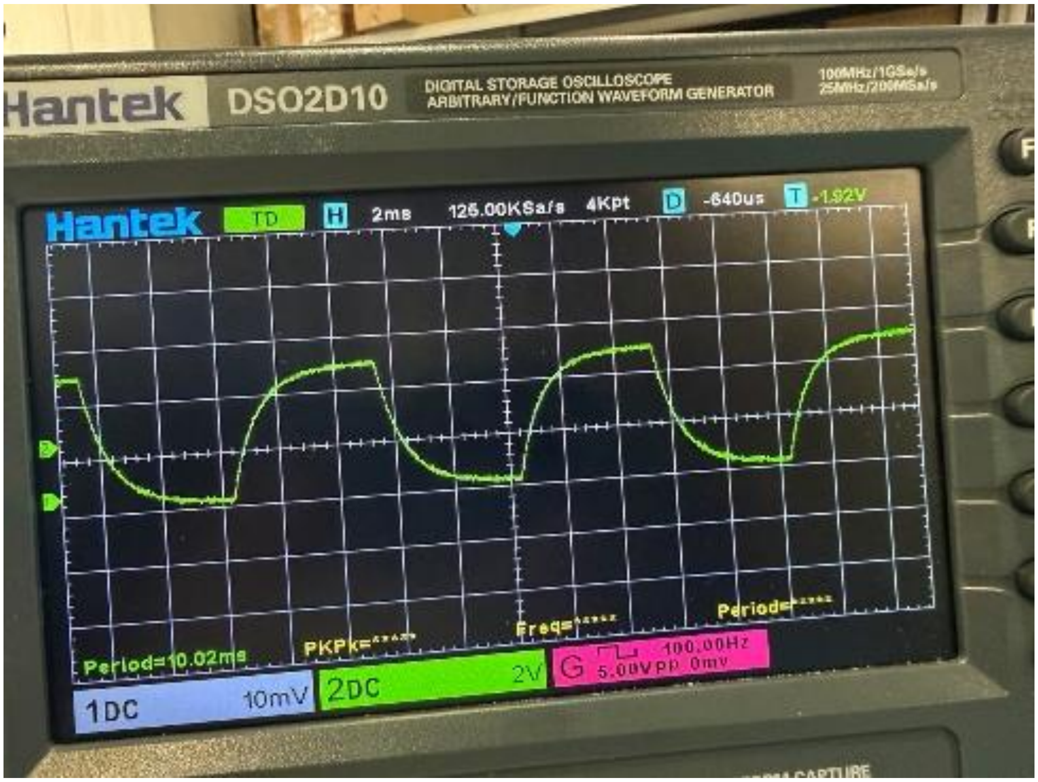
Underdamped:



Critically Damped:



Critically Damped:



Underdamped (minimal damping setting observed):